

EDUCATION

Southern Methodist University <i>Master of Interactive Technology in Digital Game Development, Software Development</i>	Dallas, Texas <i>August 2025 – May 2027</i>
Technical University of Munich <i>Exchange Student, Bachelor of Informatics</i>	Munich, Germany <i>April 2024 – August 2024</i>
National University of Singapore <i>Bachelor of Computing with Honors in Computer Science</i>	Singapore <i>August 2021 – June 2025</i>
◦ Minor: Interactive Media Development ◦ Specializations: Computer Graphics and Games, Artificial Intelligence	

EXPERIENCE

Fragcolor Pte. Ltd. <i>Game Engine Programmer Intern</i>	Singapore <i>May 2023 – October 2023</i>
◦ Developed and optimized C++ engine modules, improving usability, rendering scalability and runtime efficiency. ◦ Implemented gameplay systems for the company’s flagship title to demonstrate engine capabilities. ◦ Prototyped LLM-powered no-code workflows, reducing iteration time for designers and nontechnical users. ◦ Collaborated with engineers across time zones in a globally distributed development environment.	
The Synteraction (formerly NUS-HCI) Lab <i>Part-time Research Intern</i>	Singapore <i>February 2024 – December 2024</i>
◦ Conducted a literature review of multimedia learning systems to identify research gaps and guide new directions. ◦ Designed and led usability experiments, producing actionable insights that improved interaction flow. ◦ Developed interactive prototypes using Python and React and evaluated them through iterative user testing.	

TEAM GAME PROJECTS

<i>Hamsterballin’</i> [Shipped on Steam] <i>Lead Programmer</i>	Dallas, Texas <i>January 2026 – May 2026</i>
◦ Led 13 programmers within a 42-person team, setting technical direction and coordinating development. ◦ Presented technical updates and implementation decisions for a 3D kart racing game in Unreal Engine 5. ◦ Maintained the build pipeline, coordinated cross-team integration, and met Steam build requirements.	
<i>Out of the Dark</i> <i>General Programmer</i>	Dallas, Texas <i>September 2025 – December 2025</i>
◦ Served as the sole programmer implementing all gameplay systems for a top-down adventure game in Unity. ◦ Developed state-based enemy AI, dynamic audio, and reusable menu systems with C# scripting. ◦ Collaborated with the team to define requirements, resolve integration issues, and deliver features on schedule.	
<i>Glimmerwood</i> <i>General Programmer</i>	Singapore <i>August 2024 – December 2024</i>
◦ Served as the sole programmer implementing all gameplay systems for a side-scroller in GameMaker Studio. ◦ Developed player abilities, interactive boss encounters, and scripted gameplay events with GML.	
<i>MetalCore</i> <i>Gameplay and UI Programmer</i>	Singapore <i>January 2023 – May 2023</i>
◦ Received Best Project Award at the 22nd STePS NUS School of Computing Term Project Showcase. ◦ Developed in-game economy, upgrade systems and user interfaces in Unreal Engine 5. ◦ Collaborated with the team to design and implement combat, progression, and core gameplay loops.	

PERSONAL PROJECTS

Custom Game Engine

Dallas, Texas

C++ Programmer

August 2025 – Present

- Developed a custom 2D/3D game engine in **C++** with gameplay, physics, UI, animation, and rendering systems.
- Built **D3D11 rendering systems** for geometry, shaders, lighting, and camera controls.
- Implemented data-driven definitions, gameplay logic, console commands, and debug visualization tools.
- Developed multiple 2D/3D game projects with the engine, validating its gameplay, rendering, and tooling systems.

Stacklands

Dallas, Texas

C++ Programmer

June 2026 – July 2026

- Developed a **C++ card-based village-building game** with data-driven card definitions loaded from XML.
- Implemented card selection, dragging, stacking and perspective-aware clamping within an isometric board.
- Built a recipe system with timed interactions, sequential resource production, and ingredient consumption rules.
- Created an economy system with **weighted loot pools**, multiple draw groups, and varied card distributions.

NetChess3D

Dallas, Texas

C++ Programmer

June 2026 – July 2026

- Developed a **C++ networked 3D chess game** with complete chess rules and synchronized multiplayer matches.
- Built a non-blocking **TCP/IPv4 networking system** with WinSock 2.2, supporting client-server connections, connection states management, buffered messaging, and per-frame socket processing.
- Implemented a command protocol that validates moves locally and synchronizes gameplay actions between peers.
- Implemented normal-mapped **Blinn-Phong rendering in HLSL** with directional, point, and spot lighting.
- Added **manual OBJ loading** with XML metadata, indexed mesh generation and normal and tangent generation.

Doomenstein

Dallas, Texas

C++ Programmer

March 2026 – May 2026

- Developed a **C++ first-person shooter** with data-driven maps, actors, weapons, animations, and audio.
- Implemented raycasting, collision detection, physics-driven movement, projectile and hitscan weapons.
- Designed AI controllers, player possession, and safe actor references using **handle-based lookup**.
- Implemented per-pixel lighting, billboarded animations, 3D audio, and two-player **split-screen multiplayer**.

3D Object Reconstruction Pipeline

Singapore

Full Stack Programmer

August 2024 – May 2025

- Developed a **modular Python pipeline** for generating PBR-textured 3D assets from 2D images.
- Built containerized backend services using **Docker** and exposed the pipeline through **REST APIs**.
- Created an interactive web interface with **React and Three.js** for asset generation and 3D model preview.
- Implemented **OBJ and FBX export** for integration with **Unity and Unreal Engine** workflows.

SKILLS

Programming: C++, C#, Python, Java, C, R, Shell Scripting, PostgreSQL, HLSL, DirectX 11, OpenGL

Engines: Unreal Engine, Unity, GameMaker Studio

Tools: Git, Perforce, Visual Studio, Docker, Blender, Figma, Premier Pro, Illustrator

PUBLICATIONS

- **Sun, R.**, Li, T., Tim, T., Ng, Q., Bhojan, A. (2025). GAGA: Game Asset Generative AI Pipeline. **International Conference on Teaching, Assessment and Learning for Engineering (IEEE TALE 2025)**. - <https://doi.org/10.1109/TALE66047.2025.11346631>
- Janaka, N., Zhao, S., Ram, A., **Sun, R.**, Tan Jing Wen, S., Li, D., & Hsu, D. (2025). Progressive Sentences: Combining the Benefits of Word and Sentence Learning. **Adjunct Proceedings of the 27th International Conference on Mobile Human-Computer Interaction**, 1–12. <https://doi.org/10.1145/3737821.3749564>